

Exponential and Logarithm Functions

- $x^n = x \cdot x \cdots x$ (n times)
- $x^{\frac{1}{n}} = \sqrt[n]{x}$, $x^{m/n} = \sqrt[n]{x^m}$
- $(xy)^n = x^n y^n$

Laws of exponents

1. $x^m x^n = x^{m+n}$
2. $x^m / x^n = x^{m-n}$
3. $x^{-n} = \frac{1}{x^n}$
4. $x^0 = 1$ ($x \neq 0$)
5. $(x^m)^n = x^{mn}$

Logarithms

- $\log_a x = b \Leftrightarrow x = a^b$ ($a > 0$)
- $a^{\log_a x} = x$, $\log_a a^x = x$
- $\log_a b = \frac{\log_c b}{\log_c a}$
- $a^{\log_c b} = b^{\log_c a}$

Laws of logarithms (和指數的 5 個性質相對應)

1. $\log_a xy = \log_a x + \log_a y$
2. $\log_a \frac{x}{y} = \log_a x - \log_a y$
3. $\log_a \frac{1}{x} = -\log_a x$
4. $\log_a 1 = 0$
5. $\log_a x^n = n \log_a x$

Natural logarithms

- $e := \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = \lim_{a \rightarrow 0} (1 + a)^{\frac{1}{a}} \approx 2.7182818284$
- $\ln x := \log_e x$

$\ln x$ 就是對數，所以有以下對數性質。

- $e^{\ln x} = x, \quad \ln e^x = x$ (反函數性質)
- $\log_a b = \frac{\ln b}{\ln a}$
- $a^{\ln b} = b^{\ln a}$

Laws of natural logarithms

1. $\ln xy = \ln x + \ln y$

2. $\ln \frac{x}{y} = \ln x - \ln y$

3. $\ln \frac{1}{x} = -\ln x$

4. $\ln 1 = 0$

5. $\ln x^n = n \ln x$

Derivatives

- $\frac{d}{dx} e^x = e^x$

Proof.

$$\frac{d}{dx} e^x = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h} = e^x \cdot \lim_{h \rightarrow 0} \frac{e^h - 1}{h}$$

以下計算 $\lim_{h \rightarrow 0} \frac{e^h - 1}{h}$ 值，透過計算其倒數之極限值來計算。

令 $a = e^h - 1$ ，則有 $h = \ln(1 + a)$ 。另一方面，當 $h \rightarrow 0$ ，有 $a \rightarrow 0$ 。於是

$$\lim_{h \rightarrow 0} \frac{h}{e^h - 1} = \lim_{a \rightarrow 0} \frac{\ln(1 + a)}{a} = \lim_{a \rightarrow 0} \ln(1 + a)^{\frac{1}{a}} = \ln \left(\lim_{a \rightarrow 0} (1 + a)^{\frac{1}{a}} \right) = \ln e = 1$$

其中第三個等式是因為 $\ln$ 是連續函數，所以外面取極限相當於裡面先取極限再放入 $\ln$ 。到這裡我們就有了 $\lim_{h \rightarrow 0} \frac{h}{e^h - 1} = 1$ ，所以

$$\frac{d}{dx} e^x = e^x \cdot \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = e^x \cdot \lim_{h \rightarrow 0} \left(\frac{h}{e^h - 1} \right)^{-1} = e^x$$

□

- $\frac{d}{dx} \ln x = \frac{1}{x}$

Proof. 由 $e^{\ln x} = x$ 兩邊微分得 (用到連鎖律)

$$e^{\ln x} \cdot \frac{d}{dx} \ln x = 1$$

於是 $\frac{d}{dx} \ln x = \frac{1}{e^{\ln x}} = \frac{1}{x}$ 。

□

- $\frac{d}{dx}a^x = a^x \ln a$

Proof.

$$\frac{d}{dx}a^x = \frac{d}{dx}e^{x \ln a} = e^{x \ln a} \cdot \ln a = a^x \ln a$$

□

- $\frac{d}{dx} \log_a x = \frac{1}{x \ln a}$

Proof.

$$\frac{d}{dx} \log_a x = \frac{d}{dx} \frac{\ln x}{\ln a} = \frac{1}{\ln a} \cdot \frac{1}{x} = \frac{1}{x \ln a}$$

□